

Paper Manager pack — π _RL (Oct 5)

Paper: π _RL: Online RL Fine-tuning for Flow-based Vision-Language-Action Models

arXiv: <https://arxiv.org/abs/2510.25889>

Code: <https://github.com/RLinf/RLinf> · HF: <https://huggingface.co/RLinf>

PDF: /workspace/papers/pi-rl-2510.25889.pdf

Session: Oct 5 · due 9:30 AM

Schedule: Reviewer=3, Empiricist=4; Stakeholder/Archaeologist/Visionary empty; **Kanta not seated** — all five packs still included

Sources: Summarizer + Senior Research (/workspace/pi-rl-quiet/DELIVERABLE.md)

1. Paper summary

Online RL fine-tune for **flow-matching VLAs** ($\pi_0/\pi_{0.5}$; GR00T in appendix) so PPO can run despite intractable flow log-likelihoods and deterministic ODE sampling.

Two methods

- **Flow-Noise:** learnable noise → Gaussian denoise MDP; discard noise after FT
- **Flow-SDE:** ODE → SDE + two-layer MDP + hybrid sampler (~2× faster; often preferred)

Recipe: few-shot SFT → freeze VLM → PPO/GAE on ~300M action expert. Table1 avg **+27–31 pp** over SFT; few-shot $\pi_{0.5}$ +RL LIBERO **98.3%** > full SFT **96.9%**. OOD helps visual/env more than novel tasks (ML45 fails). Real2Sim2Real: SFT fail → RL **40%**.

BEHAVIOR bridge: OpenPI Comet **cites** this then **rejects** online RL for BEHAVIOR (sample efficiency + GPU heterogeneity) → uses **RFT**. π _RL = when fast parallel sim exists; RFT = pragmatic default for slow BEHAVIOR sims.

2. Keywords

3. Table 1 headline (ID avg %)

Model	Method	Avg	Δ vs SFT
π_0	SFT	51.1	—
π_0	Flow-SDE	**78.7**	**+27.6**
π_0	Flow-Noise	**80.3**	**+29.2**
$\pi_{0.5}$	SFT	55.6	—
$\pi_{0.5}$	Flow-SDE	**86.6**	**+31.0**
$\pi_{0.5}$	Flow-Noise	**84.7**	**+29.1**

Also: LIBERO Long 43.9 $\rightarrow$ 94.0; ManiSkill OOD $\sim$ 26 $\rightarrow$ 49–53; CALVIN Len-5 $\rightarrow$ 87; PPO>GRPO $\sim$ 6pp; GR00T App 52.5 $\rightarrow$ 89.9.

4. Role packs

Stakeholder (18+3)

Open adapter for OpenPI via RLinf; few-shot+RL can beat full SFT on reported %. Needs H100-class parallel sim+rewards; not true real online RL. Prefer π _RL with fast sim; **RFT** when BEHAVIOR-slow.

Scientific Reviewer (8+3) — MEATY (seat weight 3)

Strengths: Real gap (flow VLAs blocked PG); two mechanisms with clear ancestry; multi-benchmark + careful OOD split; open code; honest novel-task failure.

Actionable

1. Report env-steps / GPU-hours to plateau vs “beats full SFT”
2. Ablate unfreeze last VLM layers (frozen VLM may cap visual OOD)
3. When to pick Noise vs SDE (wins differ by suite)
4. “Exact likelihood” is of augmented denoise MDP, not marginal ODE policy — say so
5. Thin real n (40%) — more seeds/tasks

6. Head-to-head vs FPO / RFT under matched budgets
7. Don't overclaim venue (ICML-style keywords; acceptance TENTATIVE)

Verdict: Strong same-task mastery + partial env-shift; don't overread as open-world discovery.

Empiricist (11+3) — MEATY (seat weight 4)

Ruthless smoke: defer full online RL.

Priority	Experiment
P0 Y	Released ckpt eval; eval-only K×a grids; unit-test Flow-SDE transitions
P1 Partial	Offline/advantage-weighted CFM; critic warm-up; tiny PPO ≤8 envs ≤20–40 updates
Defer N	Tab.1 multi-bench, ManiSkill 4k, Real2Sim2Real, BEHAVIOR online RL

H1: mid noise $a \approx 0.5$ maximizes usable train-mode SR

H2: $K \in \{4, 5\}$ operating point; diminishing returns by ~8

H4: freeze VLM + expert update = ID refinement, not novel-task invention

H5: mid horizon + replan > cargo-cult H=50 during RL

Reproduce: OpenPI SFT → RLinf freeze VLM → `flow_sde` (default) or `flow_noise` → monitor train/eval gap, clip fraction, EV, KL.

One-liner: parallel sparse-success sim → near-saturated specialists; else stay on RFT.

Env: `/workspace/pi-rl-quiet/env_outline/`.

Archaeologist (11+3)

Priors: $\pi_0/\pi_{0.5}$, Lipman FM, ReinFlow, Flow-GRPO, DPPO, PPO/GAE, RLinf, Song SDE, 3DGS.

Subsequent: Comet cites then uses RFT; BEHAVIOR'25 Pi0.5 forks different post-train.

Arc: flow VLAs blocked PG → π _RL opens it → slow-sim challenges route around via RFT.

Visionary (11+3) — 2026 BEHAVIOR

	Online RL (π _RL)	RFT (Comet)
Needs	$\log \pi$ + exploration	success filter only
Sample efficiency	low	higher for slow sims
Ceiling	can exceed demos	capped by reachable modes

BEHAVIOR'25	cited, not used	Q 0.224 → 0.345
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2026 bets: RFT then short π_{RL} burst if faster eval arrives; don't freeze VLM forever; tune H' for advantages; dual $\pi_{0.5}$ +GR00T adapters.

One-liner: π_{RL} = open online half; RFT = pragmatic offline half — compose them. Skip Discord/Drive.

5. Control note

Flow policy as stochastic MDP via Flow-SDE; hybrid SDE+ODE sampler; PPO on action expert — **not** classical planner. Chunk length = credit-assignment knob.

6. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	π_{RL} vs RFT product choice
Reviewer	**8+3**	**7 actionable critiques (HIGH)**
Empiricist	**11+3**	**Smoke-feasible grid + tattoo numbers (HIGH)**
Archaeologist	11+3	FM VLAs → π_{RL} → RFT fork
Visionary	11+3	Compose RL + RFT for 2026

Spine: unlock PPO on flow VLAs; BEHAVIOR still prefers RFT. PDF for Mac. All five roles.

GitHub title (Kanta not seated): [ECE 605] - π_{RL} Making Club Pack

Paths

Artifact	Path
This pack	`/workspace/pi-rl-quiet/KANTA_PACK.md`
Senior Research	`/workspace/pi-rl-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/pi-rl-2510.25889-summary.md`
Paper PDF	`/workspace/papers/pi-rl-2510.25889.pdf`